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Solved Examples:

Examples for Chapter 5

A Demonstration Example in OR Tutor:

Fundamental Insight

Interactive Procedures in IOR Tutorial:

Interactive Graphical Method
Enter or Revise a General Linear Programming Model
Set Up for the Simplex Method—Interactive Only
Solve Interactively by the Simplex Method

Automatic Procedures in IOR Tutorial:

Solve Automatically by the Simplex Method
Graphical Method and Sensitivity Analysis

Files (Chapter 3) for Solving the Wyndor Example:

Excel Files
LINGO/LINDO File
MPL/Solvers File

Glossary for Chapter 5

See Appendix 1 for documentation of the software.

PROBLEMS

The symbols to the left of some of the problems (or their parts) have the following meaning:

D: The demonstration example listed above may be helpful.
I: You can check some of your work by using procedures listed above.

An asterisk on the problem number indicates that at least a partial answer is given in the back of the book.

5.1-1.* Consider the following problem.

Maximize $Z = 3x_1 + 2x_2$,

subject to

$$2x_1 + x_2 \leq 6$$
$$x_1 + 2x_2 \leq 6$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

I **(a)** Solve this problem graphically. Identify the CPF solutions by circling them on the graph.

(b) Identify all the sets of two defining equations for this problem. For each set, solve (if a solution exists) for the corresponding corner-point solution, and classify it as a CPF solution or corner-point infeasible solution.

(c) Introduce slack variables in order to write the functional constraints in augmented form. Use these slack variables to identify the basic solution that corresponds to each corner-point solution found in part (*b*).

(d) Do the following for *each* set of two defining equations from part (*b*): Identify the indicating variable for each defining equation. Display the set of equations from part (*c*) *after* deleting these two indicating (nonbasic) variables. Then use the latter set of equations to solve for the two remaining variables (the basic variables). Compare the resulting basic solution to the corresponding basic solution obtained in part (*c*).

(e) Without executing the simplex method, use its geometric interpretation (and the objective function) to identify the path

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(sequence of CPF solutions) it would follow to reach the optimal solution. For each of these CPF solutions in turn, identify the following decisions being made for the next iteration: (*i*) which defining equation is being deleted and which is being added; (*ii*) which indicating variable is being deleted (the entering basic variable) and which is being added (the leaving basic variable).

5.1-2. Repeat Prob. 5.1-1 for the model in Prob. 3.1-6.

5.1-3. Consider the following problem.

$$\text{Maximize} \quad Z = 2x_1 + 3x_2,$$

subject to

$$\begin{aligned} -3x_1 + x_2 &\leq 1 \\ 4x_1 + 2x_2 &\leq 20 \\ 4x_1 - x_2 &\leq 10 \\ -x_1 + 2x_2 &\leq 5 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

I **(a)** Solve this problem graphically. Identify the CPF solutions by circling them on the graph.

(b) Develop a table giving each of the CPF solutions and the corresponding defining equations, BF solution, and nonbasic variables. Calculate Z for each of these solutions, and use just this information to identify the optimal solution.

(c) Develop the corresponding table for the corner-point infeasible solutions, etc. Also identify the sets of defining equations and nonbasic variables that do not yield a solution.

5.1-4. Consider the following problem.

$$\text{Maximize} \quad Z = 2x_1 - x_2 + x_3,$$

subject to

$$\begin{aligned} 3x_1 + x_2 + x_3 &\leq 60 \\ x_1 - x_2 + 2x_3 &\leq 10 \\ x_1 + x_2 - x_3 &\leq 20 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

After slack variables are introduced and then one complete iteration of the simplex method is performed, the following simplex tableau is obtained.

Iteration	Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	x_6	Right Side
1	Z	(0)	1	0	−1	3	0	2	0	20
	x_4	(1)	0	0	4	−5	1	−3	0	30
	x_1	(2)	0	1	−1	2	0	1	0	10
	x_6	(3)	0	0	2	−3	0	−1	1	10

(a) Identify the CPF solution obtained at iteration 1.

(b) Identify the constraint boundary equations that define this CPF solution.

5.1-5. Consider the three-variable linear programming problem shown in Fig. 5.2.

(a) Construct a table like Table 5.1, giving the set of defining equations for each CPF solution.

(b) What are the defining equations for the corner-point infeasible solution (6, 0, 5)?

(c) Identify one of the systems of three constraint boundary equations that yields neither a CPF solution nor a corner-point infeasible solution. Explain why this occurs for this system.

5.1-6. Consider the following problem.

$$\text{Minimize } Z = 3x_1 + 2x_2,$$

subject to

$$\begin{aligned} 2x_1 + x_2 &\geq 10 \\ -3x_1 + 2x_2 &\leq 6 \\ x_1 + x_2 &\geq 6 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0.$$

(a) Identify the 10 sets of defining equations for this problem. For each one, solve (if a solution exists) for the corresponding corner-point solution, and classify it as a CPF solution or a corner-point infeasible solution.

(b) For each corner-point solution, give the corresponding basic solution and its set of nonbasic variables.

5.1-7. Reconsider the model in Prob. 3.1-5.

(a) Identify the 15 sets of defining equations for this problem. For each one, solve (if a solution exists) for the corresponding corner-point solution, and classify it as a CPF solution or a corner-point infeasible solution.

(b) For each corner-point solution, give the corresponding basic solution and its set of nonbasic variables.

5.1-8. Each of the following statements is true under most circumstances, but not always. In each case, indicate when the statement will not be true and why.

(a) The best CPF solution is an optimal solution.

(b) An optimal solution is a CPF solution.

(c) A CPF solution is the only optimal solution if none of its adjacent CPF solutions are better (as measured by the value of the objective function).

5.1-9. Consider the original form (before augmenting) of a linear programming problem with n decision variables (each with a nonnegativity constraint) and m functional constraints. Label each of the following statements as true or false, and then justify your answer with specific references (including page citations) to material in the chapter.

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(a) If a feasible solution is optimal, it must be a CPF solution.
(b) The number of CPF solutions is at least

$$\frac{(m+n)!}{m!n!}.$$

(c) If a CPF solution has adjacent CPF solutions that are better (as measured by Z), then one of these adjacent CPF solutions must be an optimal solution.

5.1-10. Label each of the following statements about linear programming problems as true or false, and then justify your answer.
(a) If a feasible solution is optimal but not a CPF solution, then infinitely many optimal solutions exist.
(b) If the value of the objective function is equal at two different feasible points $\mathbf{x}^*$ and $\mathbf{x}^{**}$, then all points on the line segment connecting $\mathbf{x}^*$ and $\mathbf{x}^{**}$ are feasible and Z has the same value at all those points.
(c) If the problem has n variables (before augmenting), then the simultaneous solution of any set of n constraint boundary equations is a CPF solution.

5.1-11. Consider the augmented form of linear programming problems that have feasible solutions and a bounded feasible region. Label each of the following statements as true or false, and then justify your answer by referring to specific statements (with page citations) in the chapter.
(a) There must be at least one optimal solution.
(b) An optimal solution must be a BF solution.
(c) The number of BF solutions is finite.

5.1-12.* Reconsider the model in Prob. 4.6-9. Now you are given the information that the basic variables in the optimal solution are x_2 and x_3. Use this information to identify a system of three constraint boundary equations whose simultaneous solution must be this optimal solution. Then solve this system of equations to obtain this solution.

5.1-13. Reconsider Prob. 4.3-6. Now use the given information and the theory of the simplex method to identify a system of three constraint boundary equations (in x_1, x_2, x_3) whose simultaneous solution must be the optimal solution, without applying the simplex method. Solve this system of equations to find the optimal solution.

5.1-14. Consider the following problem.

$$\text{Maximize} \qquad Z = 2x_1 + 2x_2 + 3x_3,$$

subject to

$$\begin{aligned} 2x_1 + x_2 + 2x_3 &\leq 4 \\ x_1 + x_2 + x_3 &\leq 3 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

Let x_4 and x_5 be the slack variables for the respective functional constraints. Starting with these two variables as the basic variables for the initial BF solution, you now are given the information that the simplex method proceeds as follows to obtain the optimal solution in two iterations: (1) In iteration 1, the entering basic variable is x_3 and the leaving basic variable is x_4; (2) in iteration 2, the entering basic variable is x_2 and the leaving basic variable is x_5.
(a) Develop a three-dimensional drawing of the feasible region for this problem, and show the path followed by the simplex method.
(b) Give a geometric interpretation of why the simplex method followed this path.
(c) For each of the two edges of the feasible region traversed by the simplex method, give the equation of each of the two constraint boundaries on which it lies, and then give the equation of the additional constraint boundary at each endpoint.
(d) Identify the set of defining equations for each of the three CPF solutions (including the initial one) obtained by the simplex method. Use the defining equations to solve for these solutions.
(e) For each CPF solution obtained in part (*d*), give the corresponding BF solution and its set of nonbasic variables. Explain how these nonbasic variables identify the defining equations obtained in part (*d*).

5.1-15. Consider the following problem.

$$\text{Maximize} \qquad Z = 3x_1 + 4x_2 + 2x_3,$$

subject to

$$\begin{aligned} x_1 + x_2 + x_3 &\leq 20 \\ x_1 + 2x_2 + x_3 &\leq 30 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

Let x_4 and x_5 be the slack variables for the respective functional constraints. Starting with these two variables as the basic variables for the initial BF solution, you now are given the information that the simplex method proceeds as follows to obtain the optimal solution in two iterations: (1) In iteration 1, the entering basic variable is x_2 and the leaving basic variable is x_5; (2) in iteration 2, the entering basic variable is x_1 and the leaving basic variable is x_4.

Follow the instructions of Prob. 5.1-14 for this situation.

5.1-16. By inspecting Fig. 5.2, explain why Property 1*b* for CPF solutions holds for this problem if it has the following objective function.
(a) Maximize $Z = x_3$.
(b) Maximize $Z = -x_1 + 2x_3$.

5.1-17. Consider the three-variable linear programming problem shown in Fig. 5.2.
(a) Explain in geometric terms why the set of solutions satisfying any individual constraint is a convex set, as defined in Appendix 2.

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(b) Use the conclusion in part (*a*) to explain why the entire feasible region (the set of solutions that simultaneously satisfies every constraint) is a convex set.

5.1-18. Suppose that the three-variable linear programming problem given in Fig. 5.2 has the objective function

$$\text{Maximize} \qquad Z = 3x_1 + 4x_2 + 3x_3.$$

Without using the algebra of the simplex method, apply just its geometric reasoning (including choosing the edge giving the maximum rate of increase of Z) to determine and explain the path it would follow in Fig. 5.2 from the origin to the optimal solution.

5.1-19. Consider the three-variable linear programming problem shown in Fig. 5.2.

(a) Construct a table like Table 5.4, giving the indicating variable for each constraint boundary equation and original constraint.

(b) For the CPF solution (2, 4, 3) and its three adjacent CPF solutions (4, 2, 4), (0, 4, 2), and (2, 4, 0), construct a table like Table 5.5, showing the corresponding defining equations, BF solution, and nonbasic variables.

(c) Use the sets of defining equations from part (*b*) to demonstrate that (4, 2, 4), (0, 4, 2), and (2, 4, 0) are indeed adjacent to (2, 4, 3), but that none of these three CPF solutions are adjacent to each other. Then use the sets of nonbasic variables from part (*b*) to demonstrate the same thing.

5.1-20. The formula for the line passing through (2, 4, 3) and (4, 2, 4) in Fig. 5.2 can be written as

$$(2, 4, 3) + \alpha[(4, 2, 4) - (2, 4, 3)] = (2, 4, 3) + \alpha(2, -2, 1),$$

where $0 \leq \alpha \leq 1$ for just the line segment between these points. After augmenting with the slack variables x_4, x_5, x_6, x_7 for the respective functional constraints, this formula becomes

$$(2, 4, 3, 2, 0, 0, 0) + \alpha(2, -2, 1, -2, 2, 0, 0).$$

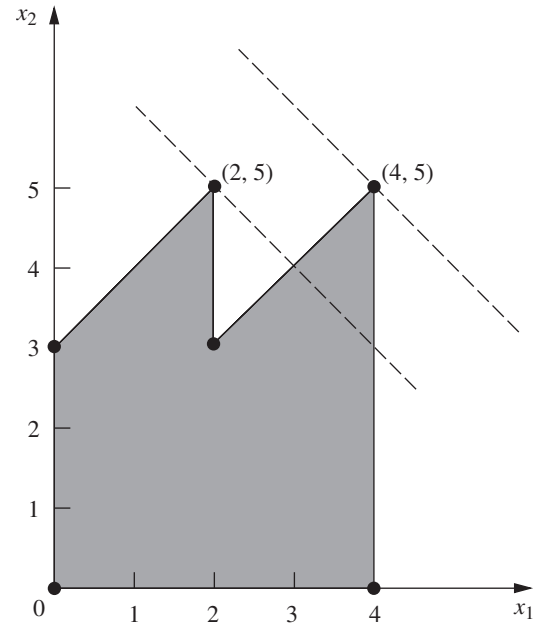
Use this formula directly to answer each of the following questions, and thereby relate the algebra and geometry of the simplex method as it goes through one iteration in moving from (2, 4, 3) to (4, 2, 4). (You are given the information that it is moving along this line segment.)

(a) What is the entering basic variable?

(b) What is the leaving basic variable?

(c) What is the new BF solution?

5.1-21. Consider a two-variable mathematical programming problem that has the feasible region shown on the graph, where the six dots correspond to CPF solutions. The problem has a linear objective function, and the two dashed lines are objective function lines passing through the optimal solution (4, 5) and the second-best CPF solution (2, 5). Note that the nonoptimal solution (2, 5) is better than both of its adjacent CPF solutions, which violates Property 3 in Sec. 5.1 for CPF solutions in linear programming. Demonstrate that this problem *cannot* be a linear programming problem by constructing the feasible region that would result if the six line segments on the boundary were constraint boundaries for linear programming constraints.



5.2-1. Consider the following problem.

$$\text{Maximize} \qquad Z = 8x_1 + 4x_2 + 6x_3 + 3x_4 + 9x_5,$$

subject to

$$\begin{aligned} x_1 + 2x_2 + 3x_3 + 3x_4 \qquad &\leq 180 \qquad \text{(resource 1)} \\ 4x_1 + 3x_2 + 2x_3 + x_4 + x_5 &\leq 270 \qquad \text{(resource 2)} \\ x_1 + 3x_2 \qquad + x_4 + 3x_5 &\leq 180 \qquad \text{(resource 3)} \end{aligned}$$

and

$$x_j \geq 0, \qquad j = 1, \ldots, 5.$$

You are given the facts that the basic variables in the optimal solution are x_3, x_1, and x_5 and that

$$\begin{bmatrix} 3 & 1 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{bmatrix}^{-1} = \frac{1}{27}\begin{bmatrix} 11 & -3 & 1 \\ -6 & 9 & -3 \\ 2 & -3 & 10 \end{bmatrix}.$$

(a) Use the given information to identify the optimal solution.

(b) Use the given information to identify the shadow prices for the three resources.

I **5.2-2.*** Work through the matrix form of the simplex method step by step to solve the following problem.

$$\text{Maximize} \qquad Z = 5x_1 + 8x_2 + 7x_3 + 4x_4 + 6x_5,$$

subject to

$$\begin{aligned} 2x_1 + 3x_2 + 3x_3 + 2x_4 + 2x_5 &\leq 20 \\ 3x_1 + 5x_2 + 4x_3 + 2x_4 + 4x_5 &\leq 30 \end{aligned}$$

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and

$$x_j \geq 0, \qquad j = 1, 2, 3, 4, 5.$$

5.2-3. Reconsider Prob. 5.1-1. For the sequence of CPF solutions identified in part (*e*), construct the basis matrix **B** for each of the corresponding BF solutions. For each one, invert **B** manually, use this $\mathbf{B}^{-1}$ to calculate the current solution, and then perform the next iteration (or demonstrate that the current solution is optimal).

I **5.2-4.** Work through the matrix form of the simplex method step by step to solve the model given in Prob. 4.1-5.

I **5.2-5.** Work through the matrix form of the simplex method step by step to solve the model given in Prob. 4.7-6.

D **5.3-1.*** Consider the following problem.

$$\text{Maximize} \qquad Z = x_1 - x_2 + 2x_3,$$

subject to

$$\begin{aligned} 2x_1 - 2x_2 + 3x_3 &\leq 5 \\ x_1 + x_2 - x_3 &\leq 3 \\ x_1 - x_2 + x_3 &\leq 2 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

Let x_4, x_5, and x_6 denote the slack variables for the respective constraints. After you apply the simplex method, a portion of the final simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	x_6	Right Side
Z	(0)	1				1	1	0	
x_2	(1)	0				1	3	0	
x_6	(2)	0				0	1	1	
x_3	(3)	0				1	2	0	

(a) Use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the final simplex tableau. Show your calculations.

(b) Identify the defining equations of the CPF solution corresponding to the optimal BF solution in the final simplex tableau.

D **5.3-2.** Consider the following problem.

$$\text{Maximize} \qquad Z = 4x_1 + 3x_2 + x_3 + 2x_4,$$

subject to

$$\begin{aligned} 4x_1 + 2x_2 + x_3 + x_4 &\leq 5 \\ 3x_1 + x_2 + 2x_3 + x_4 &\leq 4 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0, \qquad x_4 \geq 0.$$

Let x_5 and x_6 denote the slack variables for the respective constraints. After you apply the simplex method, a portion of the final simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	x_6	Right Side
Z	(0)	1					1	1	
x_2	(1)	0					1	−1	
x_4	(2)	0					−1	2	

(a) Use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the final simplex tableau. Show your calculations.

(b) Identify the defining equations of the CPF solution corresponding to the optimal BF solution in the final simplex tableau.

D **5.3-3.** Consider the following problem.

$$\text{Maximize} \qquad Z = 6x_1 + x_2 + 2x_3,$$

subject to

$$\begin{aligned} 2x_1 + 2x_2 + \frac{1}{2}x_3 &\leq 2 \\ -4x_1 - 2x_2 - \frac{3}{2}x_3 &\leq 3 \\ x_1 + 2x_2 + \frac{1}{2}x_3 &\leq 1 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

Let x_4, x_5, and x_6 denote the slack variables for the respective constraints. After you apply the simplex method, a portion of the final simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	x_6	Right Side
Z	(0)	1				2	0	2	
x_5	(1)	0				1	1	2	
x_3	(2)	0				−2	0	4	
x_1	(3)	0				1	0	−1	

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Use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the final simplex tableau. Show your calculations.

D **5.3-4.** Consider the following problem.

$$\text{Maximize} \quad Z = 20x_1 + 6x_2 + 8x_3,$$

subject to

$$\begin{aligned} 8x_1 + 2x_2 + 3x_3 &\le 200 \\ 4x_1 + 3x_2 \qquad &\le 100 \\ 2x_1 \qquad + \ x_3 &\le 50 \\ x_3 &\le 20 \end{aligned}$$

and

$$x_1 \ge 0, \qquad x_2 \ge 0, \qquad x_3 \ge 0.$$

Let x_4, x_5, x_6, and x_7 denote the slack variables for the first through fourth constraints, respectively. Suppose that after some number of iterations of the simplex method, a portion of the current simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	x_6	x_7	Right Side
Z	(0)	1				$\frac{9}{4}$	$\frac{1}{2}$	0	0	
x_1	(1)	0				$\frac{3}{16}$	$-\frac{1}{8}$	0	0	
x_2	(2)	0				$-\frac{1}{4}$	$\frac{1}{2}$	0	0	
x_6	(3)	0				$-\frac{3}{8}$	$\frac{1}{4}$	1	0	
x_7	(4)	0				0	0	0	1	

(a) Use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the current simplex tableau. Show your calculations.
(b) Indicate which of these missing numbers would be generated by the matrix form of the simplex method to perform the next iteration.
(c) Identify the defining equations of the CPF solution corresponding to the BF solution in the current simplex tableau.

D **5.3-5.** Consider the following problem.

$$\text{Maximize} \quad Z = c_1x_1 + c_2x_2 + c_3x_3,$$

subject to

$$\begin{aligned} x_1 + 2x_2 + \ x_3 &\le b \\ 2x_1 + \ x_2 + 3x_3 &\le 2b \end{aligned}$$

and

$$x_1 \ge 0, \qquad x_2 \ge 0, \qquad x_3 \ge 0.$$

Note that values have not been assigned to the coefficients in the objective function (c_1, c_2, c_3), and that the only specification for the right-hand side of the functional constraints is that the second one ($2b$) be twice as large as the first (b).

Now suppose that your boss has inserted her best estimate of the values of c_1, c_2, c_3, and b without informing you and then has run the simplex method. You are given the resulting final simplex tableau below (where x_4 and x_5 are the slack variables for the respective functional constraints), but you are unable to read the value of Z^*.

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	x_5	Right Side
Z	(0)	1	$\frac{7}{10}$	0	0	$\frac{3}{5}$	$\frac{4}{5}$	Z^*
x_2	(1)	0	$\frac{1}{5}$	1	0	$\frac{3}{5}$	$-\frac{1}{5}$	1
x_3	(2)	0	$\frac{3}{5}$	0	1	$-\frac{1}{5}$	$\frac{2}{5}$	3

(a) Use the fundamental insight presented in Sec. 5.3 to identify the value of (c_1, c_2, c_3) that was used.
(b) Use the fundamental insight presented in Sec. 5.3 to identify the value of b that was used.
(c) Calculate the value of Z^* in two ways, where one way uses your results from part (a) and the other way uses your result from part (b). Show your two methods for finding Z^*.

5.3-6. For iteration 2 of the example in Sec. 5.3, the following expression was shown:

$$\text{Final row } 0 = [-3, \quad -5 \,\vdots\, 0, \quad 0, \quad 0 \,\vdots\, 0]$$

$$+ [0, \ \tfrac{3}{2}, \ 1] \left[\begin{array}{cc:ccc:c} 1 & 0 & 1 & 0 & 0 & 4 \\ 0 & 2 & 0 & 1 & 0 & 12 \\ 3 & 2 & 0 & 0 & 1 & 18 \end{array}\right].$$

Derive this expression by combining the algebraic operations (in matrix form) for iterations 1 and 2 that affect row 0.

5.3-7. Most of the description of the fundamental insight presented in Sec. 5.3 assumes that the problem is in our standard form. Now consider each of the following other forms, where the additional adjustments in the initialization step are those presented in Sec. 4.6, including the use of artificial variables and the Big M method where appropriate. Describe the resulting adjustments in the fundamental insight.
(a) Equality constraints
(b) Functional constraints in $\ge$ form
(c) Negative right-hand sides
(d) Variables allowed to be negative (with no lower bound)

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5.3-8. Reconsider the model in Prob. 4.6-5. Use artificial variables and the Big M method to construct the complete first simplex tableau for the simplex method, and then identify the columns that will contain $\mathbf{S}^*$ for applying the fundamental insight in the final tableau. Explain why these are the appropriate columns.

5.3-9. Consider the following problem.

$$\text{Minimize} \qquad Z = 2x_1 + 3x_2 + 2x_3,$$

subject to

$$\begin{aligned} x_1 + 4x_2 + 2x_3 &\geq 8 \\ 3x_1 + 2x_2 \qquad &\geq 6 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

Let x_4 and x_6 be the surplus variables for the first and second constraints, respectively. Let $\bar{x}_5$ and $\bar{x}_7$ be the corresponding artificial variables. After you make the adjustments described in Sec. 4.6 for this model form when using the Big M method, the initial simplex tableau ready to apply the simplex method is as follows:

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	$\bar{x}_5$	x_6	$\bar{x}_7$	Right Side
Z	(0)	-1	$-4M+2$	$-6M+3$	$-2M+2$	M	0	M	0	$-14M$
$\bar{x}_5$	(1)	0	1	4	2	-1	1	0	0	8
$\bar{x}_7$	(2)	0	3	2	0	0	0	-1	1	6

After you apply the simplex method, a portion of the final simplex tableau is as follows:

Basic Variable	Eq.	Coefficient of: Z	x_1	x_2	x_3	x_4	$\bar{x}_5$	x_6	$\bar{x}_7$	Right Side
Z	(0)	-1					$M-0.5$		$M-0.5$	
x_2	(1)	0					0.3		-0.1	
x_1	(2)	0					-0.2		0.4	

(a) Based on the above tableaux, use the fundamental insight presented in Sec. 5.3 to identify the missing numbers in the final simplex tableau. Show your calculations.

(b) Examine the mathematical logic presented in Sec. 5.3 to validate the fundamental insight (see the $\mathbf{T}^* = \mathbf{MT}$ and $\mathbf{t}^* = \mathbf{t} + \mathbf{vT}$ equations and the subsequent derivations of $\mathbf{M}$ and $\mathbf{v}$). This logic assumes that the original model fits our standard form, whereas the current problem does not fit this form. Show how, with minor adjustments, this same logic applies to the current problem when $\mathbf{t}$ is row 0 and $\mathbf{T}$ is rows 1 and 2 in the initial simplex tableau given above. Derive $\mathbf{M}$ and $\mathbf{v}$ for this problem.

(c) When you apply the $\mathbf{t}^* = \mathbf{t} + \mathbf{vT}$ equation, another option is to use $\mathbf{t} = [2, 3, 2, 0, M, 0, M, 0]$, which is the *preliminary* row 0 before the algebraic elimination of the nonzero coefficients of the initial basic variables $\bar{x}_5$ and $\bar{x}_7$. Repeat part (*b*) for this equation with this new $\mathbf{t}$. After you derive the new $\mathbf{v}$, show that this equation yields the same final row 0 for this problem as the equation derived in part (*b*).

(d) Identify the defining equations of the CPF solution corresponding to the optimal BF solution in the final simplex tableau.

5.3-10. Consider the following problem.

$$\text{Maximize} \qquad Z = 3x_1 + 7x_2 + 2x_3,$$

subject to

$$\begin{aligned} -2x_1 + 2x_2 + x_3 &\leq 10 \\ 3x_1 + \ \ x_2 - x_3 &\leq 20 \end{aligned}$$

and

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad x_3 \geq 0.$$

You are given the fact that the basic variables in the optimal solution are x_1 and x_3.

(a) Introduce slack variables, and then use the given information to find the optimal solution directly by Gaussian elimination.

(b) Extend the work in part (*a*) to find the shadow prices.

(c) Use the given information to identify the defining equations of the optimal CPF solution, and then solve these equations to obtain the optimal solution.

(d) Construct the basis matrix $\mathbf{B}$ for the optimal BF solution, invert $\mathbf{B}$ manually, and then use this $\mathbf{B}^{-1}$ to solve for the optimal solution and the shadow prices $\mathbf{y}^*$. Then apply the optimality test for the matrix form of the simplex method to verify that this solution is optimal.

(e) Given $\mathbf{B}^{-1}$ and $\mathbf{y}^*$ from part (*d*), use the fundamental insight presented in Sec. 5.3 to construct the complete final simplex tableau.

5.4-1. Consider the model given in Prob. 5.2-2. Let x_6 and x_7 be the slack variables for the first and second constraints, respectively. You are given the information that x_2 is the entering basic variable and x_7 is the leaving basic variable for the first iteration of the simplex method and then x_4 is the entering basic variable and x_6 is the leaving basic variable for the second (final) iteration. Use the procedure presented in Sec. 5.4 for updating $\mathbf{B}^{-1}$ from one iteration to the next to find $\mathbf{B}^{-1}$ after the first iteration and then after the second iteration.

I **5.4-2.*** Work through the revised simplex method step by step to solve the model given in Prob. 4.3-4.

I **5.4-3.** Work through the revised simplex method step by step to solve the model given in Prob. 4.7-5.

I **5.4-4.** Work through the revised simplex method step by step to solve the model given in Prob. 3.1-6.

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